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The Role of Organic and Biological Fertilizers on the Growth, Yield of Mung Bean Plant and Water Use Efficiency in Water Deficit Conditions in Northern Iran

Poorya Sadeghi Kochsafhani, Seyed Amirabbas Mousavi Mirkalaei , Morteza Moballeghi , Morteza Samdeliri , and Maryam Akbarpour 

Department of Agriculture, Chalous Branch, Islamic Azad University, Chalous, Iran

ABSTRACT

Knowledge of the yield of short-season crops under insufficient irrigation is of particular importance for regions that face water deficit during the plant growth period. This experiment was carried out by investigating the effects of organic and biological fertilizers on the growth, yield and water use efficiency of mung bean plants under water deficit conditions and the traits affecting grain yield. Experiment treatments including irrigation at three levels (100, 80 and 60% of water requirement (WR)), organic fertilizer (0 and 16 and 24 t/ha manure) and biofertilizer (control, Arbuscular mycorrhizal (AM) fungi and enterobacter (EB)). In stress and non-stressed conditions, the application of AM and organic fertilizer has increased the number of seeds and the number of plant pods. The highest weight of one hundred seeds was obtained in the condition of 80% WR and application of organic and biological fertilizer. The number of primary branches was higher with the use of EB, but the highest number of secondary branches was observed in the plant with AM. The highest plant height, biological yield and seed were observed in the condition of 100% WR and 24 t/ha of organic fertilizer and application of AM and EB. The highest water use efficiency was obtained with irrigation of 80% WR and application of AM and 24 t/ha of organic fertilizer, as well as irrigation of 60% WR and application of AM and 24 t/ha of organic fertilizer. The use of organic fertilizer along with biofertilizers in stress conditions has increased the measured traits, and this increase was more in severe stress with AM than EB fertilizer. The results of the path analysis showed that the weight of one hundred seeds, the number of seeds in the pod and biological yield are the most direct effects on seed yield.

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Introduction

Legumes, as one of the plant sources rich in protein, are considered the most important source of human food after cereals, which are cultivated all over the world and have adapted to different weather conditions from moderate to hot and from wet to dry (Meena et al., 2018). These plants, with the biological stabilization of nitrogen (N) while improving soil fertility, are effective in preventing soil erosion as cover plants or in rotation with many crops and play an important role in the stability of agricultural systems (Romanyà and Casals 2020; Yuvaraj, Pandiyan, and Gayathri 2020).

Mung bean (*Vigna radiata* (L.) Wilczek) is a warm season pulse legume. It is currently cultivated on about six million hectares worldwide, most of which are located in Asia (FAO 2022). As a legume, the crop fixes most of its own N requirement and contributes significantly to improving the sustainability of farming systems (Chauhan and Williams 2018; Ebert 2014; Peoples et al. 2009).

Today, due to the high nutritional benefits of mung bean, the demand for this product has increased, but due to its low yield and its growth period in hot seasons and lack of water, farmers do not have much desire to cultivate it. The average annual long-term rainfall in northern Iran is over 1000 mm, and more than 93% of it falls from November to May. Therefore, irrigation during mung bean growing season is needed to maintain and increase crop growth and yield. Water deficit is a critical issue that limits mung bean growth by affecting anatomical, morphological, physiological and biochemical processes (Ambachew, Alamirew, and Melese 2014; Sadiq, Akram, and Ashraf 2018).

Mycorrhizal symbiosis is a useful strategy in helping plants cope with adverse environmental conditions (Rabie 2005). Arbuscular mycorrhizal (AM) reduce drought stress by directly absorbing water and transferring to the host plant through fungal hyphae, and on the other hand, it increases the absorption of nutrients and the rate of net photosynthesis in the host plant (Badr et al. 2020). AM increases soil water retention capacity and prevents nutrients from leaching by affecting the stability of soil aggregates (He et al. 2021). Hashem et al. (2019) reported that drought stress reduced the relative water content and membrane stability index in pea plants, while the treatment of biochar and AM individually or in combination reduced the destructive effects of drought stress to the extent possible. They decreased attention and increased the amount of these traits in normal irrigation.

Growth promoting bacteria are part of the rhizosphere bacteria that improve plant growth and yield and can increase the secretion of auxin hormone under drought stress conditions and then increase the effective level of plant roots and finally improve the absorption of water and nutrients by the plant. It is cultivated. Enterobacters (EB) are growth-promoting bacteria that naturally exist in water, soil, and decaying materials (Brook and Ledney 1994).

Mazandaran province is the region where rice is most commonly cultivated, and a considerable amount of water is allocated to its production. Due to the proximity of mung bean planting date with rice, farmers are facing the problem of water shortage for mung bean production. On the other hand, for the continuation of production in sustainable agricultural systems, it is focused on the need to develop technologies and methods that do not have a negative effect on the environment and lead to improved economic efficiency and reducing the effects of insufficient irrigation on plants in order to maintain agricultural jobs. The use of organic and biological fertilizers due to the improvement of soil organic matter, biomass and soil microbial activity and increasing water retention can be a suitable alternative to continue the production of agricultural products, especially in drought stress conditions (Ansari and Mahmood 2017; Irfan et al. 2019; Mirbakhsh 2022). By examining different sources, it was found that there is little information about the effect of combining two biofertilizers (EB and AM) with animal manure on mung bean in low irrigation conditions. Considering the role of biofertilizers in stimulating growth and the positive role of organic fertilizers in conserving water and improving growth conditions as well as the lack of water in the mung bean growing season in Mazandaran province of Iran, this study aims to investigate the effect of combining biofertilizer and animal manure on It was done on reducing the effects of lack of irrigation in mung bean production.

Methods and materials

The location and characteristics of the study site soil

This research was carried out in the research farm of Islamic Azad University Faculty of Agriculture, Chalus branch, Mazandaran Province, Iran 40° 58'N and 53° 69' E with an altitude of 3 m above sea level during two crop years 2018–19 and 2019–2020. Composite sampling was done from the depth of 30 cm of the soil of the experiment site was carried out, and the amount of some physical and chemical characteristics of the soil were measured. The sampling of the soil in the area is shown in Table 1.

Table 1. Physical and chemical characteristics of the study site soil.

Texture	Phosphorus (ppm)	Potassium (ppm)	Total N (%)	Organic carbon (%)	pH	EC (ds/m)
clay loam	8.8	341	0.12	1.4	7.06	1.24

Experimental design and treatments

The experiment was carried out in the form of split-factorial in the form of a basic randomized complete block design with 3 replications. The main factor includes irrigation at three levels (100, 80 and 60% of water requirement (WR)) and organic fertilizer (0 (nonuse), 16 and 24 t/ha of cow manure) and biological fertilizer (control, arbuscular mycorrhizal (AM) fungi and *Enterobacter* (EB)). The species of AM used in this experiment was *Glomus mosseae*, which was obtained from Tehran Water and Soil Research Institute, Iran. The seeds of the NM-92 variety (*Vigna radiata* (L.) Wilczek) were inoculated with AM before sowing.

The nitrozist biofertilizer that contains EB was obtained from Royan Tisan Sabze Co. The bacterial population density in all inoculums was 108 spore/g. For each 1 kg of seeds, 10 ml of 20% sugar solution was added to the seeds and mixed well inside a plastic bag until the seeds became completely moist and sticky. Then, about 15 g of fungi powder and 300 ml of liquid nitrozist were added to the seeds and mixed again. The seeds were spread in the shade for about 20 minutes and after drying, they were planted in the field.

Agricultural operations

Land preparation operations, including plowing, disking and leveling, were carried out before planting. The land preparation operation was carried out at the end of June. The plowing operation was done with a chisel plow, and then the vertical disk was stirred twice and leveled. Then plots with dimensions of 2.5 × 6 m and ridges with a depth of 20 cm and a distance between rows of 50 cm were created. The distance between the plots in each block was considered to be 2 rows of planting (1 m) and the distance between the blocks was 2 m.

The amount of water consumed in the full irrigation treatment (100% WR) was calculated as the amount of water needed to complete the field capacity at a depth of 60 cm in the soil profile. The amount of irrigation water used in the 80% and 60% water treatments was calculated compared to the full irrigation treatment. The required water was supplied and irrigated by a pump motor with a flow rate of 6 l/s, separately and using a canal. Measurement of soil-water content for all treatments was done one day before irrigation by gravimetric method. Plants were irrigated when 40% of the available soil moisture was consumed at a depth of 60 cm in the full irrigation treatment. Irrigation intervals varied from 10 days in the early stage to 6 days in the middle stage of the season. Irrigation was completely done until the stage of 4–5 leaves in all treatments. The plants were irrigated 10 times on average during the growing season. In each irrigation, about 55.24 mm of water was used for complete irrigation.

Measured traits

At the time of harvesting (after removing the two side rows as a marginal effect), 10 plants were selected from each plot and plant height, the number of primary and secondary branches of each plant, the number of pods per plant, the number of seeds per pod, and the weight of 100 seeds were calculated. Also, plants were harvested from each plot with an area of 1.5 m² and after drying the samples of each plot, its biological yield was recorded. Then, after separating the seeds from the harvested pods, the seed yield of each plot per unit area was calculated. Water use efficiency was calculated by using of following formula (Şahin, Akçalı, and Tiryakioğlu 2024):

$$WUE = \frac{Y}{W}$$

WUE = water use efficiency (kg/m³)

Y = total yield (kg)

W = applied irrigation water (m³)

Data analysis

Data were analyzed using the GLM procedure of SAS version of 9.2 (SAS 2008) software. The comparison of means was done with the least significant difference (LSD) method at the 5% level. To ensure the uniformity of the variances in order to integrate the data, Bartlett's test was performed, and the results indicated the non-significance of the data and composite analysis was performed.

Results and discussion

Plant height

The results of variance analysis showed that the quadruple effect of year × irrigation × organic fertilizer × biofertilizer was significant on plant height (Appendix Table A1). The highest value of plant height was achieved with 100% WR and application of EB and 24 t/ha of organic fertilizer in the first and second years at the rate of 46.95 and 44.2 cm, respectively. The lowest value of plant height was recorded when the plant was irrigated with 60% of its water requirements without organic fertilizer in the first year, and with EB and no biofertilizer, at 26.13 and 26.28 cm, respectively (Table 2). The plant height of the mung bean has decreased with the increase of irrigation interval. In conditions of severe stress, the use of AM has worked better than EB fertilizer and has increased the height of the plant. It seems that the activity of AM was more in absorbing water and nutrients. The hydraulic conductivity of the root system in AM plants is higher than that of non-AM plants, which is due to the increase of the root surface or the length of the AM roots. Also, organic fertilizer has been able to reduce the adverse effects of drought stress and prevented a large decrease in plant height. The decrease in the height of the plant due to the increase in the irrigation cycle could be considered as a result of the decrease in the ability of the plant to absorb nutrients and synthesize and transport nutrients due to the lack of water, which caused a decrease in the growth and accumulation of plant

Table 2. Effect of irrigation, manure and biological fertilizer on plant height of mung bean.

year	Irrigation	biologic	manure		
			0	16 (t/ha)	24 (t/ha)
First year	100%	EB	39.23 ±3.27 hijk	42.13 ±3.24 cde	44.2 ±4.02 b
		AM	38.23 ±2.55 jklm	41.36 ±2.95 cdefg	42.81 ±3.29 bc
		CO	35.08 ±2.7 pqr	37.62 ±3.58 klmn	41.46 ±3.46 cdefg
	80%	EB	32.8 ±2.73 stu	41.18 ±3.17 cdefg	37.67 ±3.42 klmn
		AM	36.13 ±2.41 nopq	36.13 ±2.58 nopq	40.47 ±3.11 defgh
		CO	29.43 ±2.26 wx	37.87 ±3.61 klmn	37.15 ±3.1 lmno
	60%	EB	26.13 ±2.18 y	31.93 ±2.46 uv	34.23 ±3.11 rs
		AM	29.7 ±1.98 wx	34.9 ±2.49 pqr	35.47 ±2.73 opqr
		CO	26.28 ±2.02 y	32.1 ±3.06 tuv	31.11 ±2.59 uvw
Second year	100%	EB	42.12 ±3.51 cde	44.25 ±3.4 b	46.95 ±4.27 a
		AM	37.45 ±2.5 klmn	40.37 ±2.88 efgh	42.36 ±3.26 c
		CO	36.68 ±2.82 mnop	39.13 ±3.73 hijk	40.3 ±3.36 fghi
	80%	EB	33.85 ±2.82 rst	39.82 ±3.06 ghij	38.53 ±3.5 ijkl
		AM	37 ±2.47 lmno	42.25 ±3.02 cd	41.78 ±3.21 cdef
		CO	30.42 ±2.34 vwxx	36.63 ±3.49 mnop	38.03 ±3.17 jklm
	60%	EB	27 ±2.25 y	36.2 ±2.78 nopq	32.89 ±2.99 stu
		AM	30.78 ±2.05 vwxx	38.23 ±2.73 jklm	34.25 ±2.63 rs
		CO	29.24 ±2.25 x	34.56 ±3.29 qrs	31.65 ±2.64 uv

Means in each column followed by similar letter(s) are not significantly different at 5% probability level, using LSD Test.

dry matter. The decrease in the production of photosynthetic materials due to water stress has been reported by other researchers (Maghsoudi, Emam, and Pessarakli 2016; Nazar et al. 2015). Also, Li et al. (2022) stated that the decrease in plant height in water deficit conditions was due to the decrease in internode length or the number of internodes.

In this research, the irrigation cycle of 60% WR was reduced once due to the decrease in the ability of the plant to absorb nutrients and the effect of water stress on the growth enzymes activity of the mung bean plant. In addition to limiting the growth of cells, the lack of water caused the plant to enter the reproductive stage earlier, as a result, the height of the first pod decreased. Also, Gölgül, Kırnak, and Ali İrik (2023) presented a similar report about the effect of water deficit on the plant height of mung bean. The application of organic fertilizer has reduced the effects of drought stress. By improving the cationic exchange capacity of the soil, organic and biological fertilizers caused better and easier plant access to water and nutrients, especially N, and, as a result, increased vegetative growth and plant height.

Number of primary and secondary branches

The number of primary and secondary branches was affected by the effect of year $\times$ irrigation $\times$ organic fertilizer $\times$ biological fertilizer (Appendix Table A1). The results of the effect of the number of primary and secondary branches are shown in Tables 3 and 4. Based on the results, it was observed that the most primary branches occurred under irrigation conditions of 100% WR, 16 t/ha of organic fertilizer, and EB application in the second year (4.85 branches) and the application of 16 t/ha of organic fertilizer and without biological fertilizer during the second year under irrigation conditions of 100% WR (4.63 branches) (Table 3). The highest number of secondary branches in the irrigation conditions of 100% WR and 24 t/ha of organic fertilizer and AM in the second year at the rate of 4.66 branches and then in the irrigation condition of 100% WR and 16 t/ha of organic fertilizer and AM in the first year (4.5 branches) were observed. The lowest number of primary and secondary branches was also observed in the irrigation condition of 60% WR and without organic and biological fertilizers in both years (Table 4). Lack of water has led to a decrease in the number of secondary branches. This result is consistent with the report of Mazhar et al. (2023) who stated that lack of water has an effect on the number of branches of mung bean plant. Bhattacharya (2021) stated that the lack of available water for the plant caused a decrease in growth and the observation of such a thing was due to the reduction

Table 3. Effect of irrigation, manure and biological fertilizer on number of primary branches of mung bean.

year	Irrigation	biologic	manure		
			0	16 (t/ha)	24 (t/ha)
First year	100%	EB	4.00 $\pm$ 0.33 hi	4.58 $\pm$ 0.38 bc	3.65 $\pm$ 0.33 mno
		AM	3.2 $\pm$ 0.25 rs	3.9 $\pm$ 0.28 jk	4.13 $\pm$ 0.34 fg
		CO	3.23 $\pm$ 0.22 qr	4.53 $\pm$ 0.41 c	3.90 $\pm$ 0.3 jk
	80%	EB	3.15 $\pm$ 0.26 st	3.20 $\pm$ 0.27 rs	4.00 $\pm$ 0.36 hi
		AM	3.63 $\pm$ 0.28 no	3.60 $\pm$ 0.26 no	3.85 $\pm$ 0.32 k
		CO	3.74 $\pm$ 0.25 l	3.20 $\pm$ 0.29 rs	3.9 $\pm$ 0.3 jk
	60%	EB	3.03 $\pm$ 0.25 u	2.80 $\pm$ 0.23 x	3.64 $\pm$ 0.33 mno
		AM	2.96 $\pm$ 0.23 uv	4.07 $\pm$ 0.29 gh	3.74 $\pm$ 0.31 l
		CO	2.72 $\pm$ 0.18 y	3.13 $\pm$ 0.28 st	2.96 $\pm$ 0.23 uv
	Second year	EB	3.71 $\pm$ 0.31 lm	4.85 $\pm$ 0.4 a	4.23 $\pm$ 0.39 de
		AM	3.66 $\pm$ 0.28 mn	4.16 $\pm$ 0.3 ef	4.26 $\pm$ 0.36 d
		CO	3.14 $\pm$ 0.21 st	4.63 $\pm$ 0.42 b	3.96 $\pm$ 0.31 ij
		EB	3.3 $\pm$ 0.28 pq	3.75 $\pm$ 0.31 l	4.09 $\pm$ 0.37 fg
		AM	3.74 $\pm$ 0.29 l	3.71 $\pm$ 0.27 lm	3.91 $\pm$ 0.33 jk
		CO	3.3 $\pm$ 0.22 pq	3.58 $\pm$ 0.33 o	3.89 $\pm$ 0.3 jk
	60%	EB	3.13 $\pm$ 0.26 st	2.89 $\pm$ 0.24 vv	3.12 $\pm$ 0.28 t
		AM	2.87 $\pm$ 0.22 wx	3.26 $\pm$ 0.23 qr	3.36 $\pm$ 0.28 p
		CO	2.65 $\pm$ 0.18 y	2.86 $\pm$ 0.26 wx	2.98 $\pm$ 0.23 u

Means in each column followed by similar letter(s) are not significantly different at 5% probability level, using LSD Test.

Table 4. Effect of irrigation, manure and biological fertilizer on number of secondary branches of mung bean.

year	Irrigation	biologic	manure		
			0	16 (t/ha)	24 (t/ha)
First year	100%	EB	3.63 ± 0.3 gh	4.50 ± 0.38 b	4.1 ± 0.29 cd
		AM	2.12 ± 0.16 pq	2.94 ± 0.21 i	3.67 ± 0.33 fgh
		CO	2.00 ± 0.31 qr	3.61 ± 0.3 h	3.66 ± 0.14 fgh
	80%	EB	2.29 ± 0.18 mno	2.32 ± 0.17 mn	2.53 ± 0.23 k
		AM	1.9 ± 0.16 rst	1.73 ± 0.14 vwx	1.30 ± 0.09 AB
		CO	1.7 ± 0.13 wxy	1.79 ± 0.13 tuvw	2.19 ± 0.2 nop
	60%	EB	1.07 ± 0.09 C	2.63 ± 0.22 jk	1.52 ± 0.11 z
		AM	0.91 ± 0.07 D	2.24 ± 0.16 nop	1.88 ± 0.17 rstu
		CO	0.81 ± 0.07 D	1.65 ± 0.14 xyz	1.15 ± 0.08 C
Second year	100%	EB	4.05 ± 0.31 d	4.20 ± 0.3 c	4.66 ± 0.42 a
		AM	2.16 ± 0.18 op	3.03 ± 0.25 i	3.80 ± 0.27 e
		CO	2.3 ± 0.18 mn	3.75 ± 0.27 efg	3.77 ± 0.34 ef
	80%	EB	2.39 ± 0.2 lm	2.00 ± 0.17 qr	2.52 ± 0.18 kl
		AM	1.32 ± 0.1 A	1.78 ± 0.13 tuvwx	1.96 ± 0.18 rs
		CO	1.77 ± 0.15 uvwx	1.83 ± 0.15 stuv	2.23 ± 0.16 nop
	60%	EB	1.10 ± 0.09 C	2.56 ± 0.18 k	1.58 ± 0.14 yz
		AM	1.18 ± 0.1 BC	2.73 ± 0.23 j	1.93 ± 0.14 rs
		CO	0.84 ± 0.06 D	2.32 ± 0.17 mn	1.28 ± 0.12 AB

Means in each column followed by similar letter(s) are not significantly different at 5% probability level, using LSD Test.

of photosynthesis, the reduction of minerals and the change in the amount of plant hormones due to the lack of water. Also, Abdelaal (2015) reported on the decrease in the number of primary shoots of mung bean in the condition of water deficit and stated that the stress of water deficit reduced the supply of cultivated materials and caused a decrease in the number of shoots by disrupting the process of absorption and transfer of food elements. In this study, the plant under 60% WR condition was subjected to severe stress of water deficit, which caused disturbance in the amount of photosynthesis, transpiration and metabolic processes of the plant, and as a result, the plant did not have the ability to produce more branches due to the reduction of dry matter. In treatments containing biofertilizers, especially those treated with AM, the water conductivity per root length unit is two to three times higher in host plants (Hardie and Leyton 1981). This factor makes the AM plants able to absorb water and nutrients including nitrogen and phosphorus more effectively from the soil and thus helps the plant to grow better in water deficit conditions.

Number of pods per plant

The results of analysis of variance showed that the quadruple effect of year × irrigation × organic fertilizer × biological fertilizer on the number of pods per plant was significant (Appendix Table A1). The highest number of pods per plant in the condition of 100% WR and the application of EB with the application of 24 and 16 t/ha of organic fertilizer in the second year was observed at the rate of 44 and 44 pods, respectively. The lowest number of pods per plant was also observed in the condition of irrigation of 60% WR and without organic and biological fertilizers in the second year to 24 pods (Table 5). In moderate stress, the use of EB fertilizer had a higher number of pods than AM, but in severe stress conditions, AM had a higher number of pods than the control and EB. In general, increasing the interval of irrigation has reduced the number of pods. In their study, Saleh et al. (2018) and Amanullah et al. (2022) reported a decrease in the number of pods in a mung bean plant under increased irrigation intervals. Other researchers' findings also indicate a significant decrease in the number of pods per plant due to moisture stress (Mazhar et al. 2023; Nawaz et al. 2021). It seems that one of the reasons for the decrease in the number of pods per plant in the irrigation cycle of 60% WR is the decrease in the growth period of the plant, as a result of which the production of photosynthetic substances decreases. The decrease in the synthesis of photosynthetic materials and the resulting

Table 5. Effect of irrigation, manure and biological fertilizer on number of pods in plant of mung bean.

year	Irrigation	biologic	manure		
			0	16 (t/ha)	24 (t/ha)
First year	100%	EB	38.23 ± 2.94 f	43.30 ± 2.89 a	40.20 ± 2.87 de
		AM	34.00 ± 2.13 no	35.36 ± 3.21 jkl	40.33 ± 4.03 cde
		CO	31.23 ± 2.84 qr	37.25 ± 3.39 fgh	36.34 ± 2.8 hij
	80%	EB	37.53 ± 2.89 fg	34.91 ± 2.33 klmn	37.58 ± 2.68 fg
		AM	31.85 ± 1.99 pq	36.2 ± 3.29 hij	31.23 ± 3.12 qr
		CO	26.64 ± 2.42 wx	32.36 ± 2.94 p	31.32 ± 2.41 pqr
	60%	EB	25.63 ± 1.97 x	32.12 ± 2.14 pq	30.36 ± 2.17 rs
		AM	29.36 ± 1.84 stu	33.50 ± 3.05 o	32.25 ± 3.23 pq
		CO	28.3 ± 2.57 uv	31.65 ± 2.88 pq	29.16 ± 2.24 tu
Second year	100%	EB	41.41 ± 3.19 bc	44.2 ± 2.95 a	44.23 ± 3.16 a
		AM	37.86 ± 2.37 fg	36.25 ± 3.3 hij	41.65 ± 4.17 b
		CO	29.85 ± 2.71 st	43.30 ± 3.94 a	37.52 ± 2.89 fg
	80%	EB	35.68 ± 2.74 jk	35.97 ± 2.4 ijk	41.20 ± 2.94 bcd
		AM	29.35 ± 1.83 stu	34.35 ± 3.12 lmno	39.52 ± 3.95 e
		CO	27.72 ± 2.52 vw	35.49 ± 3.23 jk	35.60 ± 2.74 jk
	60%	EB	28.36 ± 2.18 uv	34.23 ± 2.28 mno	31.25 ± 2.23 pqr
		AM	36.33 ± 2.27 hij	35.32 ± 3.21 jklm	36.80 ± 3.68 ghi
		CO	24.36 ± 2.21 y	30.21 ± 2.75 rst	31.16 ± 2.4 qr

Means in each column followed by similar letter(s) are not significantly different at 5% probability level, using LSD Test.

increase in intra-plant competition, along with the shedding of flowers due to water deficit stress, have caused a decrease in the number of pods in the plant. The high number of pods in the plant during the 100% WR, the presence of sufficient moisture in the stages of flowering and the beginning of the formation of pods, caused more vegetative growth and the formation of a large number of pods.

Biofertilizers improve growth by increasing water absorption by plant roots and increasing plant resistance to environmental stresses (Basu, Rabara, and Negi 2017; Bhardwaj et al. 2014). Also, studies have shown that AM increased the net growth rate and dry matter accumulation of mung bean plants under stress conditions (Habibzadeh et al. 2013; Turan 2021), as a result of which the number of pods per plant is higher in treatments containing AM under stress conditions.

Number of seeds in a pod

The results of analysis of variance showed that the number of seeds in the pod was influenced by the effect of year × irrigation × organic fertilizer × biological fertilizer (Appendix Table A1). The highest number of seeds in the pod was observed in the condition of 100% WR and the application of 24 t/ha of organic fertilizer and AM in the second year at the rate of 9.85 seeds. The lowest number of pods per plant was observed in the condition of irrigation of 60% WR and without organic and biological fertilizers in the first year (7 seeds) (Table 6). Increasing the interval of irrigation has reduced the number of pods, but the use of organic and biological fertilizers has played an effective role in reducing the effects of drought stress. In the conditions of 60% WR, the use of 24 and 16 t/ha of organic fertilizer along with AM had the highest number of seeds in the pod. Namvar et al. (2011) stated that the number of seeds in a pod is more influenced by genetic factors than environmental conditions, but environmental conditions such as water, nutrients and the length of the growth period also affect that trait, and the lack of any of the above factors causes a decrease. The number of seeds in the pod is (seed abortion). In this study, it was also shown that the lack of water caused a relatively small decrease in the number of seeds in mung bean pods. This result is consistent with the study of Mazhar et al. (2023).

The use of organic and biological fertilizers has increased the number of seeds in mung bean pods in stressed and non-stressed conditions. Wahid et al. (2019) stated that the use of AM can reduce the effects of stress and improve the balance of nutrients between soil and plants. It seems that the special structural features of organic fertilizer, in terms of having many voids, help to improve the soil

Table 6. Effect of irrigation, manure and biological fertilizer on number of seeds in a pod of mung bean.

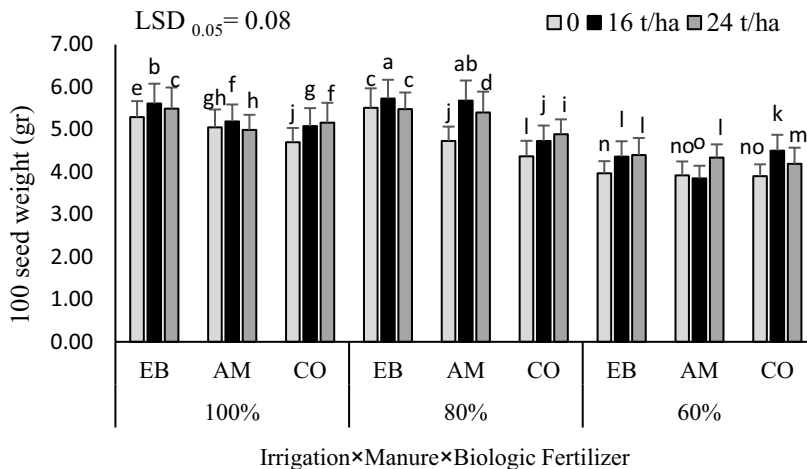
year	Irrigation	biologic	manure		
			0	16 (t/ha)	24 (t/ha)
First year	100%	EB	9.33 ± 1.04 de	9.43 ± 0.86 bcd	9.61 ± 0.96 bc
		AM	8.60 ± 0.78 ijkl	7.54 ± 0.58 uv	9.07 ± 0.76 fg
		CO	8.40 ± 0.65 lmno	8.76 ± 0.97 hij	8.80 ± 0.8 hi
	80%	EB	7.87 ± 0.87 rst	8.37 ± 0.76 mno	8.50 ± 0.85 klmn
		AM	7.60 ± 0.69 uv	9.15 ± 0.7 ef	8.20 ± 0.68 op
		CO	6.98 ± 0.54 yz	8.42 ± 0.94 lmn	8.35 ± 0.76 no
	60%	EB	7.32 ± 0.81 w	8.00 ± 0.73 pqrs	7.40 ± 0.82 a
		AM	7.57 ± 0.69 uv	8.12 ± 0.63 pq	8.02 ± 0.67 pqr
		CO	7.00 ± 0.54 yz	7.25 ± 0.81 wx	7.26 ± 0.66 wx
Second year	100%	EB	9.36 ± 1.04 d	9.42 ± 0.86 cd	9.63 ± 0.96 b
		AM	9.15 ± 0.83 ef	9.35 ± 0.72 de	9.85 ± 0.82 a
		CO	8.56 ± 0.66 jklm	8.39 ± 0.93 mno	8.94 ± 0.81 gh
	80%	EB	8.41 ± 0.93 lmn	8.64 ± 0.79 ijk	8.76 ± 0.88 hij
		AM	7.85 ± 0.71 rst	9.02 ± 0.69 fg	8.33 ± 0.69 no
		CO	6.74 ± 0.52 A	8.63 ± 0.96 ijk	8.65 ± 0.79 ijk
	60%	EB	6.8 ± 0.76 zA	7.26 ± 0.66 wx	7.93 ± 0.79 qrs
		AM	6.94 ± 0.63 yzA	7.81 ± 0.6 st	8.50 ± 0.71 klmn
		CO	6.80 ± 0.52 zA	7.70 ± 0.86 tu	7.10 ± 0.65 xy

Means in each column followed by similar letter(s) are not significantly different at 5% probability level, using LSD Test.

ventilation conditions in the long term, which is beneficial for proper plant growth and soil biological activities. By using compounds such as fungi and organic fertilizers, by preventing the wastage of nutrients and preserving nutrient ions, it has increased the efficiency of fertilizers and improved the absorption of nutrients and moisture retention (Semenov et al. 2022), and finally, it has improved plant growth and the number of seeds in the pod.

100 seed Weight

According to the variance analysis table, the triple effect of irrigation × biofertilizer × organic fertilizer on the weight of one hundred seeds was significant (Appendix Table A1). The highest weight of 100 seeds was observed under irrigation conditions of 80% WR and 16 t/ha of organic fertilizer and with the use of EB (5.73 g) and AM (5.68 g). Also, the lowest weight of 100 seeds was observed in the irrigation conditions of 60% WR and the application of 16 t/ha of organic fertilizer and AM (Figure 1). There is no significant difference between the weight of 100 seeds of mung bean under conditions of medium stress and without stress, this is the conclusion that moisture stress has little effect on the weight of 100 seeds. In this study, it seems that in the irrigation cycle of 80% WR, due to the decrease in

**Figure 1.** Effect of irrigation, manure and biological fertilizer on 100 seed weight of mung bean.

the number of seeds per plant, the stored materials in the filling stage have reached a lower number of seeds compared to the treatment of 100% WR. As a result, the seeds received and stored more carbohydrate materials, which increased the weight of 100 seeds in the 80% WR treatment. The reason for the decrease in the weight of 100 seeds in the 60% WR treatment was the occurrence of drought stress. The occurrence of drought stress, especially in the stage of pod filling, due to the limitation of the photosynthetic source, has caused a reduction in photosynthesis, materials not reaching the seeds, and also shortened the period of seed filling, and as a result, the seed size and the weight of 100 seeds have decreased. Some studies reported the decrease in the weight of mung bean plants under the stress of water deficit and stated that the weight of 100 mung beans decreases with increasing irrigation intervals (Amanullah et al. 2022; Gölgül, Kırnak, and Ali İrik 2023).

Inoculation with biological fertilizers increased the weight of 100 seeds compared to no inoculation. This increase is probably due to the conservation of water in the organic fertilizer treatments, which reduced the effects of drought stress on the transfer of photosynthetic materials during seed filling in stress conditions by absorbing more water by fungi and bacteria for the plant, as a result, the weight of one hundred seeds increased compared to the treatment of nonuse of organic and biological fertilizers.

Biological yield

The biological yield of the mung bean plant was affected by the treatments (Appendix Table A2). The highest biological yield was observed in the conditions of 100% WR and 24 t/ha of organic fertilizer and the application of AM (279.1 g/m³) and EB (275.35 g/m³) in the second year. The lowest biological yield was also observed in the irrigation conditions of 60% WR and without organic fertilizer and the application of AM in the first and second years at the rate of 110.9 and 113.5 g/m³, respectively (Table 7). Increasing the irrigation interval has reduced the biological yield. Ahmad et al. (2015), Abbasi et al. (2022) and Nawaz et al. (2021) reported similar results on mung bean plant. Robertson, Peoples, and Fukai (2004) stated that water deficit stress reduces mung bean yield by reducing the dry weight of the whole plant. It seems that the reason for the reduction of dry matter in severe stress conditions is the lack of further expansion and continuity of the leaf surface, which reduces the use of received light and as a result, the production of dry matter decreases. In severe stress conditions, treatments inoculated with AM had higher biological yield than treatments without inoculation. The difference in the yield of AM treatments and different irrigation regimes is related to the amount of

Table 7. Effect of year, irrigation, manure and biological fertilizer on biological yield of mung bean.

year	Irrigation	biologic	manure		
			0	16 (t/ha)	24 (t/ha)
First year	100%	EB	250.00 ± 22.73 c	260.3 ± 20.02 b	270.93 ± 22.58
		AM	229.30 ± 22.93 fg	242.3 ± 20.19 b	270.13 ± 27.01
		CO	215.40 ± 19.58 h	229.6 ± 17.66 d	254.46 ± 21.21
	80%	EB	246.19 ± 24.62 fg	242.31 ± 20.19 d	252.87 ± 25.29
		AM	206.85 ± 18.8 ij	224.51 ± 17.27 hi	227.73 ± 18.98
		CO	164.04 ± 16.4 l	215.60 ± 17.97 no	200.5 ± 20.05
	60%	EB	146.84 ± 13.35 rs	162.09 ± 12.47 o	197.86 ± 16.49
		AM	110.20 ± 11.02 v	144.04 ± 12 m	210.36 ± 21.04
		CO	129.02 ± 11.73 u	151.32 ± 11.64 no	200.00 ± 16.67
Second year	100%	EB	239.5 ± 23.95 ef	245.3 ± 20.44 a	275.35 ± 27.54
		AM	240.3 ± 21.85 c	261.36 ± 20.1 a	279.10 ± 23.26
		CO	222.63 ± 22.26 g	238.6 ± 19.88 e	248.30 ± 24.83
	80%	EB	231.30 ± 21.03 fg	242.30 ± 18.64 d	252.63 ± 21.05
		AM	159.40 ± 15.94 jk	222.90 ± 18.58 n	203.76 ± 20.38
		CO	201.3 ± 18.3 kl	219.20 ± 16.86 hij	226.30 ± 18.86
	60%	EB	151.29 ± 15.13 st	160.20 ± 13.35 q	184.43 ± 18.44
		AM	113.58 ± 10.33 r	165.30 ± 12.72 p	190.30 ± 15.86
		CO	133.05 ± 12.1 t	156.14 ± 12.01 q	180.30 ± 15.03

Means in each column followed by similar letter(s) are not significantly different at 5% probability level, using LSD Test.

water absorption and mineral nutrients, so that plants inoculated with AM affect the water balance of plants in drought stress and as a result due to the absorption of more water and mineral nutrients increase the yield of the product. The results of this research are consistent with the reports of other researchers (Gough et al. 2021; Kardoni et al. 2019). In the irrigation cycle with longer intervals, the use of organic fertilizer increased the biological yield compared to its nonuse. Vejan et al. (2021) stated that organic fertilizer helps to improve plant growth through selective absorption and controlled release of nutrients. In this study, it seems that the use of organic fertilizer at longer irrigation intervals in mung bean improved plant growth due to better absorption of nutrients by the plant.

Seed yield

The results of analysis of variance showed that the effect of irrigation, organic and biological fertilizer on seed yield was significant (Appendix Table A2). The results showed that the highest seed yield was observed in the conditions of irrigation of 100% WR and the application of 24 t/ha of organic fertilizer and EB and AM at the rate of 123.3 and 118.20 g/m³, respectively (Figure 2). In conditions of moderate and severe stress, the application of biofertilizer has improved the yield compared to the control. In stress conditions, AM has a better result than EB in organic fertilizer application conditions. In general, irrigation with 60% WR decreased grain yield compared to other irrigation treatments. The reason for the decrease in seed yield under irrigation conditions of 60% WR can be attributed to the lack of access of plants to irrigation water, as a result of which an increase the competition between plants for water, a decrease in yield components (number of pods per plant, number of seeds per pod, seeds weight) and an increase in the percentage of pod porosity occur. Lack of water through the reduction of the leaf surface and their fall leads to a reduction of the photosynthetic source and a decrease in the activity of enzymes, which will result in a decrease in grain yield. Islam et al. (2021) stated in a report that lack of water has reduced the yield of mung beans. At each irrigation level, 80% and 60% WR inoculated with AM was superior to the non-inoculation. If the irrigation interval is increased from 100% to 80%, the maximum amount of organic fertilizer and the use of AM can prevent the large decrease in seed yield. Habibzadeh et al. (2013) stated that treatments inoculated with mycorrhiza increased the yield of seeds. In this experiment, it seems that the increase in seed yield of AM treatments compared to the control is due to the increase in dry matter and the flow of photosynthetic materials to the productive organs. On the other hand, the use of organic fertilizer increased the seed yield due to maintaining soil moisture and absorbing nutrients and making them available to the plant during the growth period. Organic fertilizer keeps harmful salts in the soil and water in the soil (Atafar et al. 2010), and in this way, the absorption of water by the plant becomes easier and the seed yield increases during irrigation intervals.

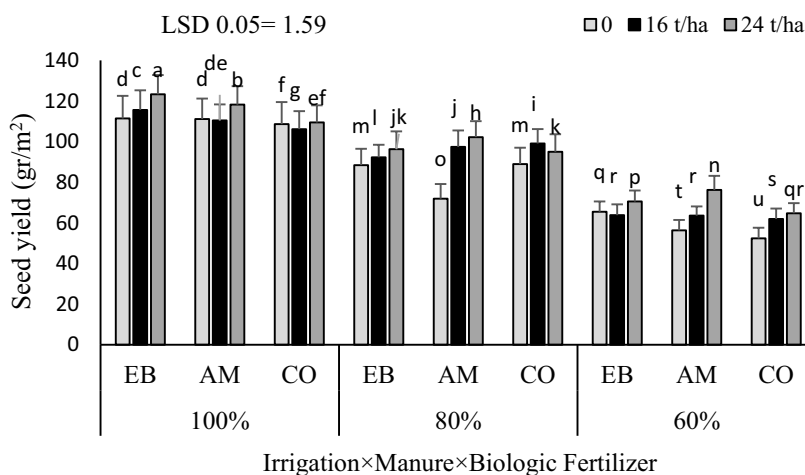


Figure 2. Effect of irrigation, manure and biological fertilizer on seed yield of mung bean.

Water use efficiency

The efficiency of water use was affected by the use of organic and biological fertilizers and low irrigation (Appendix Table A2). The highest of water use efficiency was obtained with irrigation of 80% WR and application of AM and 24 t/ha of organic fertilizer (0.277 kg/m^3), as well as irrigation of 60% WR and application of AM and 24 t/ha of organic fertilizer (0.276 kg/m^3), respectively. The lowest efficiency of water use was achieved in irrigation with 60% WR and no application of bio-fertilizer and organic fertilizer at the rate of 0.19 kg/m^3 (Figure 3). In this study, the use of organic fertilizer due to the conservation of water in the soil, in addition to preventing water wastage, caused better growth of the plant during the growth period, and as a result, the amount of dry matter and the yield of mung beans increased and the efficiency of water use increased. Wang et al. (2020) reported in their study that the use of organic fertilizer increases the efficiency of water use due to maintaining soil moisture. Also, Celik, Ortas, and Kilic (2004) and Tao et al. (2018) stated that the application of organic fertilizer increases the water holding capacity in the soil. In order to reduce moisture stress, organic fertilizers are used, which increase the water holding capacity of the soil and the water available to the plant and increase productivity. Ullah et al. (2019) stated in a study that drought stress leads to a decrease in the water use efficiency under stress conditions.

AM fungus has also increased the water use efficiency in the mung bean plant. Similar results were obtained by Habibzadeh et al. (2013), Abbasi et al. (2022), Ahmad et al. (2013), Lotfi et al. (2021) on mung bean plant. It has been reported that plants with AM symbiosis have better growth and higher yields because they absorb more nutrients and water from the soil (Lotfi et al. 2021). The increase in the absorption of water and nutrients by the root can be caused by the growth of hyphae up to 20 mm from the root surface compared to the growth of 1.5 mm of hairy roots and the greater penetration of the hyphae in contrast to the low penetration of the root into soil pores. In this study, in the 80% and 60% irrigation cycles, the coexistence of AM fungi with plant roots has led to an increase in the absorption of water and nutrients, as a result of which the plant growth improved and the seed yield increased in the treatments inoculated with AM. In this study, in the 80% and 60% irrigation cycles, the coexistence of AM fungi with plant roots has led to an increase in the absorption of water and nutrients, as a result, plant growth improved and seed yield increased in treatments inoculated with mycorrhiza. Therefore, the water use efficiency increased with the increase of seed yield.

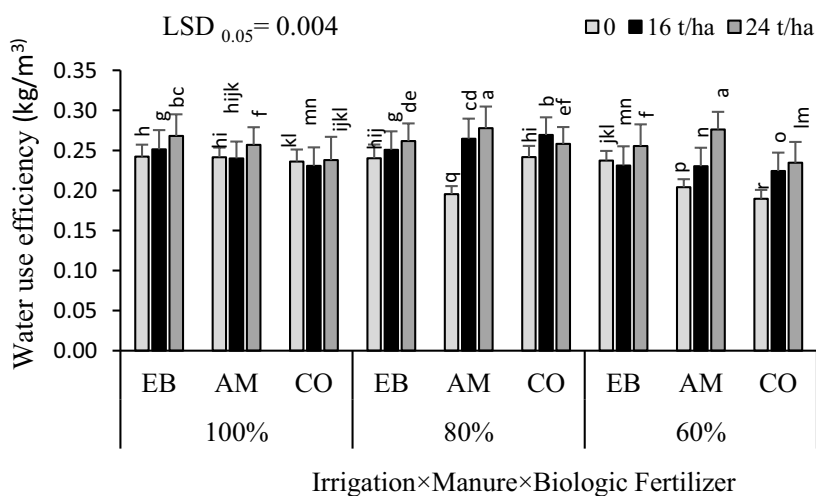


Figure 3. Effect of irrigation cycle, manure and biological fertilizer on water use efficiency.

Correlation analysis

The results of trait correlation coefficients showed a significant correlation between the traits of plant height, number of secondary branches, number of seeds in a pod, biological yield, seed protein, and water use efficiency on seed yield. The highest positive correlation coefficients were observed for the number of seeds in the pod ($r = 0.79$) and biological yield ($r = 0.76$), respectively (Figure 4). Zhou et al. (2022) reported the highest correlation of seed yield with plant height, root dry weight, number of pods per plant, number of seeds per plant and biological yield.

The reason for the high correlation coefficient of seed yield with biological yield indicates that the seed yield has increased with the increase of total biomass. Considering that the seed is the result of the photosynthetic activity of organs such as leaves. Therefore, the high correlation between these two traits is not far from expected, and it means that to have a high yield, plants with good green growth and appropriate vegetative power are needed.

Also, seed yield had a significant and negative correlation with seed protein. With the increase in seed yield, the content of seed protein has decreased. Singh and Kumar (2014) also reported a negative relationship between seed yield and seed protein content. During the seed filling period, the plant transfers storage materials including carbohydrates from the stem and leaf organs to the seed. Increasing the amount of carbohydrates in the grain causes a decrease in the protein concentration of the grain and an increase in the protein concentration of the straw.

Stepwise regression analysis

According to the obtained results, 4 traits were entered into the regression model, which are biological yield (x_8), number of seeds in the pod (x_7), weight of one hundred seeds (x_5) and stem weight (x_4). Finally, the model justified 81.1% of the changes related to the grain yield trait. According to the model, the weight of 100 seeds has the greatest effect on seed yield.

$$\text{Yield} = -45.19 + 0.335x_8 + 0.679x_7 + 5.66x_5 + 0.76x_4$$

According to Table 8, it can be seen that the biological yield trait is the first trait entered into the model and it alone justifies about 58.6% of the changes in the trait related to seed yield. The next trait that was included in the model was the number of seeds in the pod, which along with the biological yield, explains 75.6% of the changes in the trait depending on the seed yield. The third and fourth traits

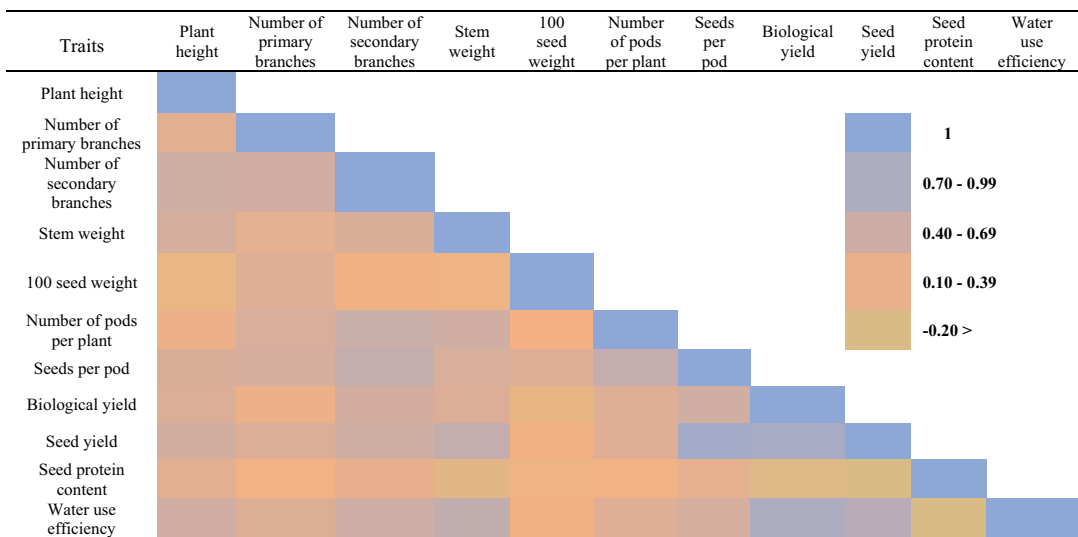


Figure 4. Correlation between measured traits.

Table 8. Results of step-by-step regression for seed yield trait as a dependent variable in mung bean.

Step	Traits	Explanation coefficient	MS regression	Regression coefficient
1	Biological yield	58.6	17278.1	0.33
2	Number of seeds in pod	75.6	5026.6	0.67
3	100 seed weight	78.7	905.7	5.6
4	Stem weight	81.1	684.3	0.76

Table 9. Path analysis of correlation between seed yield and remaining traits in step-by-step regression model.

Traits	Direct effect	Indirect effect		
		Biological yield	Number of seeds in pod	100 seed weight
Biological yield	0.032-		0.034-	0.03
Number of seeds in pod	0.205	0.0055-		0.168
100 seed weight	0.689	0.0023-	0.035-	

entered into the model were hundred seed weight and stem weight, which together with the first two traits explain about 81.1% of the changes in seed yield, which are consistent with the results of the correlation table. In general, among the examined traits, only the biological yield traits, the number of seeds per pod, the weight of one hundred seeds and the dry weight of the stalk were recognized as the most important traits affecting the amount of changes in the dependent variable (seed yield). The trait of biological yield alone has justified 58.6% of changes in seed yield. As the growth of the plant improves, the number of branching and the length of the internodes, and then the number of flowers and pods increases, and the storage time of the material in the seed increases, so the yield of the seed increases.

Path analysis

According to the regression results, the traits entered into the regression model were subjected to path analysis. In the analysis of path, seed yield was placed as a dependent variable against traits of hundred seed weight, number of seeds in a pod and biological yield as independent variables. To determine the direct and indirect effects of each of these variables with the function variable. According to the results of Table 9, the weight of one hundred seeds ($r = 0.689$), the number of seeds in a pod ($r = 0.205$) and biological yield ($r = 0.032$) are the most direct effect on seed yield, respectively. Biological yield had a negative effect on grain yield. The correlation of traits of weight of one hundred seeds, number of seeds in a pod and biological yield are 0.31, 0.22 and 0.76, respectively.

It is also observed that the direct effect of the hundred seed weight trait is greater than the correlation between this trait and grain yield, and this shows that other traits had a greater effect on seed yield through indirect effects than the direct effect of the hundred seed weight trait.

Conclusion

The results of analysis of variance showed that the effect of year, irrigation, biological and organic fertilizer on plant height, number of primary and secondary branches, number of pods and seeds, seed and biological yield, percentage and efficiency of water use were significant. Plant height, stem dry weight, seed and biological yield decreased with increasing irrigation frequency. The use of organic fertilizers had an effect on the measured traits, so that under stress conditions, the need for irrigation increased by 80% WR and the use of 16 t/ha of stem dry weight. The highest seed and biological yields were also influenced by the use of organic and biological fertilizers, so that the use of 24 t/ha of organic fertilizers along with AM in irrigation with 100% WR resulted in the highest seed yield and under medium stress conditions (80% WR). Application of 24 t/ha of organic fertilizer and EB had the greatest effect in increasing grain yield. The role of AM in increasing seed yield was proven. The use of

this fungi along with organic fertilizer can lead to an increase in seed yield. The efficiency of water use has also increased with the use of organic fertilizers and AM. In the conditions of water deficit, the use of organic fertilizer along with AM has the highest water use efficiency.

The results of trait correlation coefficients showed a significant correlation between the traits of plant height, number of secondary branches, number of seeds in a pod, biological yield, seed protein, and water use efficiency on seed yield. Also, the results of the path analysis showed that the weight of one hundred seeds, the number of seeds in a pod and biological yield are the most direct effects on seed yield.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Seyed Amirabbas Mousavi Mirkalaei  <http://orcid.org/0000-0003-3922-6130>

Morteza Moballeghi  <http://orcid.org/0000-0003-1062-1516>

Morteza Samdeliri  <http://orcid.org/0000-0002-7216-6937>

Maryam Akbarpour  <http://orcid.org/0000-0003-3635-8779>

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Appendix

Table A1. Variance analysis of growth traits and yield components of mung bean.

S.O.V	df	Plant height	No of primary branches	No of secondary branches	No of pods per plant	No of seeds in a pod	100 grain weight
Year (Y)	1	0.18	0.423	45.39	26.55	3.93*	2.33
Block(Y)	4	0.89	1.437	148.60	12.55	0.45	1.54
Irrigation (I)	2	56.70	7.396	0.02	0.04	0.00	0.00
I×Y	2	20.01**	16.00*	198.03*	47.33*	6.13*	15.42**
Error a	8	2.45	3.548	41.40	11.62	1.82	2.14
Biologic (B)	2	7.06	1.185	22.86	156.06	3.56	4.20
Manure (M)	2	2.17	3.992	475.70	263.14	0.13	1.09
B×Y	2	6.45	2.32	20.01	149.37	3.10	5.62
M×Y	2	2.01	5.2**	250.12**	349.00**	0.12	2.75
I×B	4	1.63	0.363	5.44	1.98	0.52	1.60
I×M	4	1.62	1.087	25.44	57.05	0.76	6.01
I×B×Y	4	4.2*	0.1	6.05	2.89	0.47	1.54
I×M×Y	4	3.5	3.3*	23.09	56.65**	0.79*	5.95**
B×M	4	1.72	0.234	28.58	19.01	0.51	3.01
B×M×Y	4	1.1	0.2	0.02	22.83	0.49	2.23
I×B×M	8	0.988	1.692	21.02	8.10	0.64	4.01*
I×B×M×Y	8	5.2**	3.89*	52.15*	9.19	0.58	2.17
Error b	96	1.26	0.45	19.67	18.49	0.37	1.54

* and **: significantly difference at the 0.05 and 0.01 probability levels, respectively.

Table A2. Variance analysis of yield and water use efficiency of mung bean

S.O.V	df	Biological yield	Seed yield	Water use efficiency
Year (Y)	1	316.76	1416.48	8.25*
Block(Y)	4	1055.95	10166.45	1.79
Irrigation (I)	2	2.56	2.29	1.25
I×Y	2	16664.12**	36136.99*	2.55
Error a	8	1373.57	12767.42	26.54
Biologic (B)	2	1235.17	1856.76	3.25
Manure (M)	2	8025.08	40145.87	72.01
B×Y	2	1155.52**	2791.19*	4.26
M×Y	2	7117.76**	37420.98**	61.38**
I×B	4	156.42	6200.74	1.61
I×M	4	3254.29	15243.96	1.91
I×B×Y	4	131.84	5400.58	3.42
I×M×Y	4	2670.16	15302.97	2.23
B×M	4	421.13	968.55	5.52
B×M×Y	4	326.65*	1863.67	3.06
I×B×M	8	1282.36*	2229.09	9.02*
I×B×M×Y	8	126.14	3654.21*	6.62
Error b	96	132.58	1249.63	4.35

* and **: significantly difference at the 0.05 and 0.01 probability levels, respectively.